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# Un ideal y su radical definen la misma variedad algebraica

**Ejercicio 1.** Sea $I \subset K[x_1, \ldots, x_n]$ un ideal. Demuestra que $\mathbb{V}(I) = \mathbb{V}\left(\sqrt{I}\right)$.

***Solución*:** Veamos ambas inclusiones por separado:

- $\boxed{\subset}$ Sea $a = (a_1, \ldots, a_n) \in \mathbb{V}(I)$, entonces $\forall p \in I : p(a) = 0$. Sea $q \in \sqrt{I}$, entonces $\exists m \in \mathbb{N} : q^m \in I$ y, por tanto, $0 = (q^m)(a) = (q(a))^m$. Como $K$ es un dominio de integridad, no contiene nilpotentes no nulos y, por tanto, $q(a) = 0$. Luego, $a \in \mathbb{V}(\sqrt{I})$.
- $\boxed{\supset}$ Sea $a = (a_1, \ldots, a_n) \in \mathbb{V}\left(\sqrt{I}\right)$, entonces $\forall p \in \sqrt{I} : p(a) = 0$. Como $I \subset \sqrt{I}$, tenemos que $\forall p \in I : p(a) = 0$ y, por tanto, $a \in \mathbb{V}(I)$.

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